

ABOUT OUR TEAM

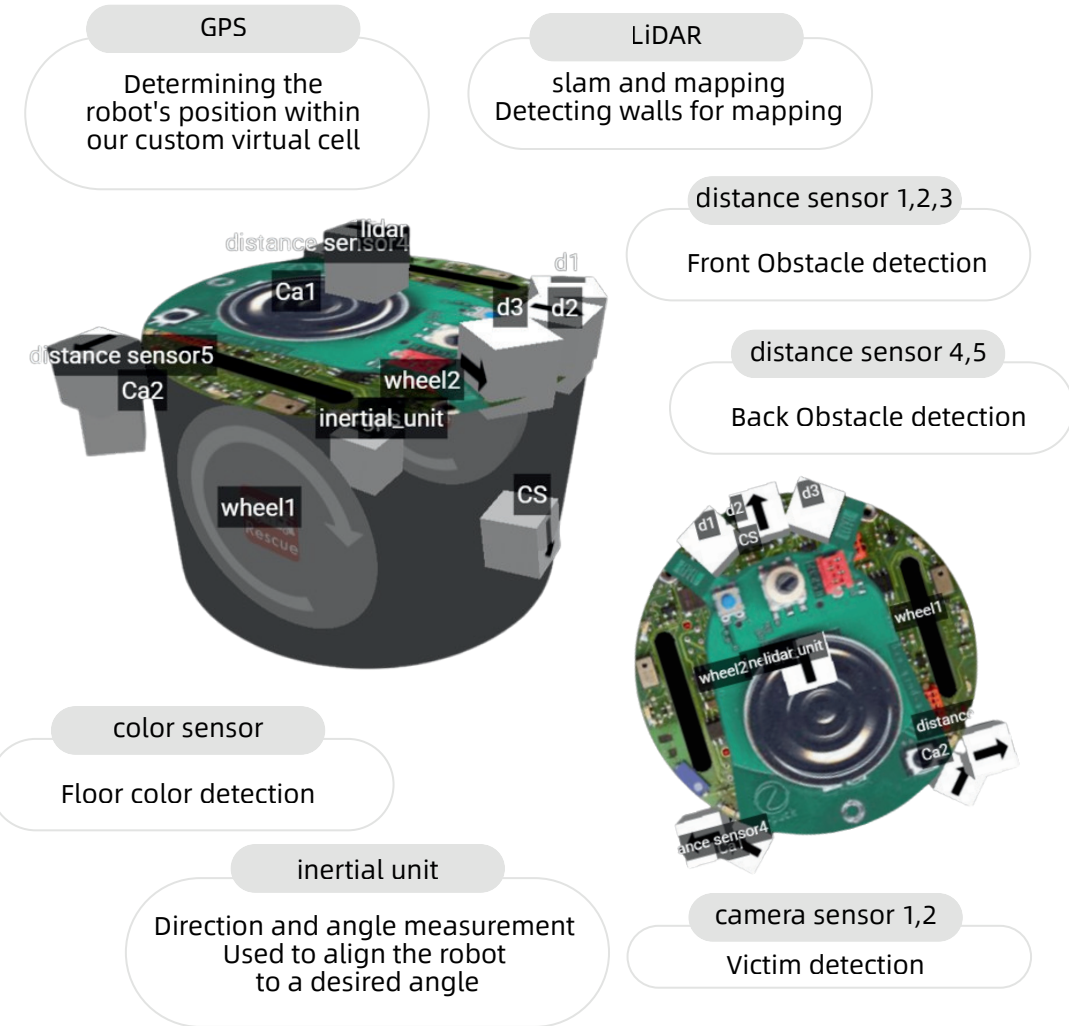


- Naeun Kim - Team Leader & Lead Developer
- Nahyun Kim - Navigation & SLAM Engineer
- Jihee Jung - AI & Computer Vision Engineer
- Seojun Yang - Research & Testing Engineer
- Mingeun Kim - COACH

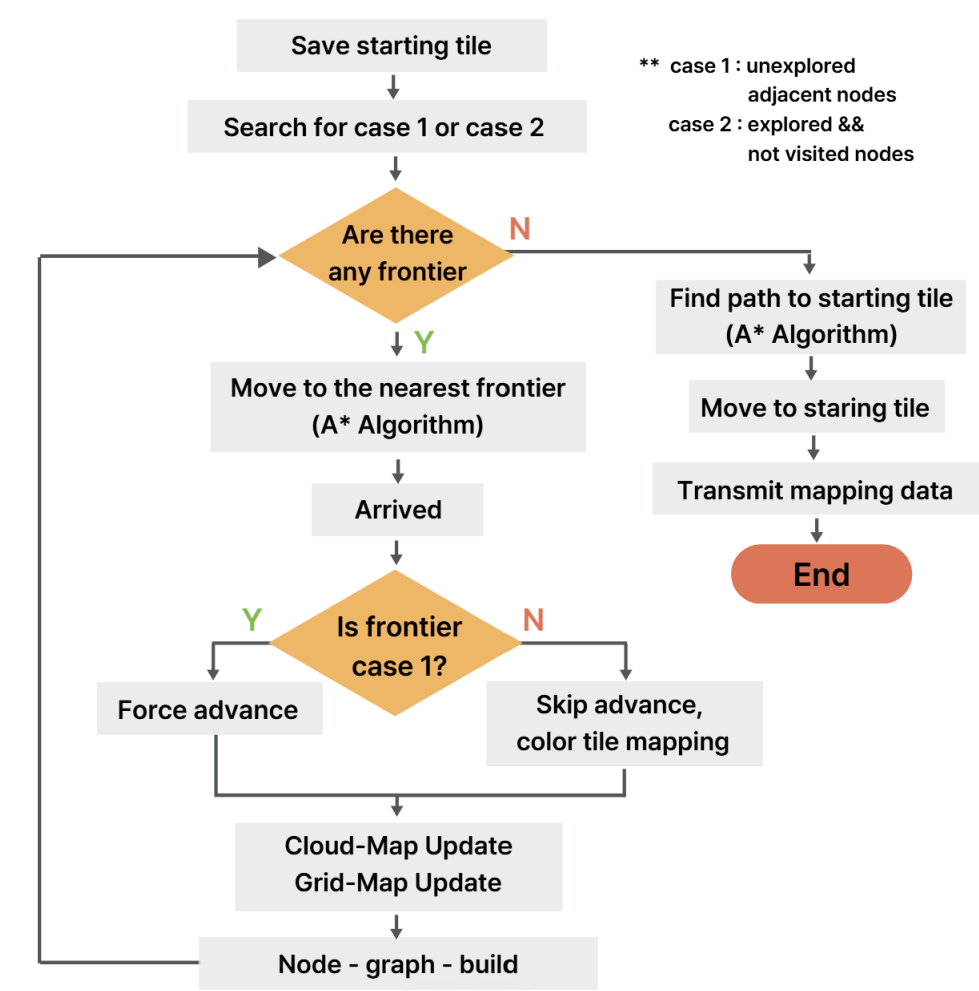
 Awards: 2026 Robocup Korea Open 1st in Rescue Simultaion

We used Python as our development language, and the total preparation period for the competition was approximately 13 weeks.

ROBOT INTRODUCTION



NAVIGATION & MAPPING



Autonomous Exploration Strategy

Environment Representation The robot utilizes a grid-based mapping approach, dividing the environment into 12 cm cells. Each cell maintains real-time updates via:

- LiDAR-based Cloud Map: Monitors wall occupancy and structural obstacles.
- Color Sensors: Detects floor conditions to identify traversable surfaces.
- Status Tracking: Categorizes exploration status into Explored (region entered) and Visited (center of the cell reached).

Path Planning & Exploration Strategy

The system constructs a topological graph by treating cell boundaries as nodes and connecting adjacent nodes as edges.

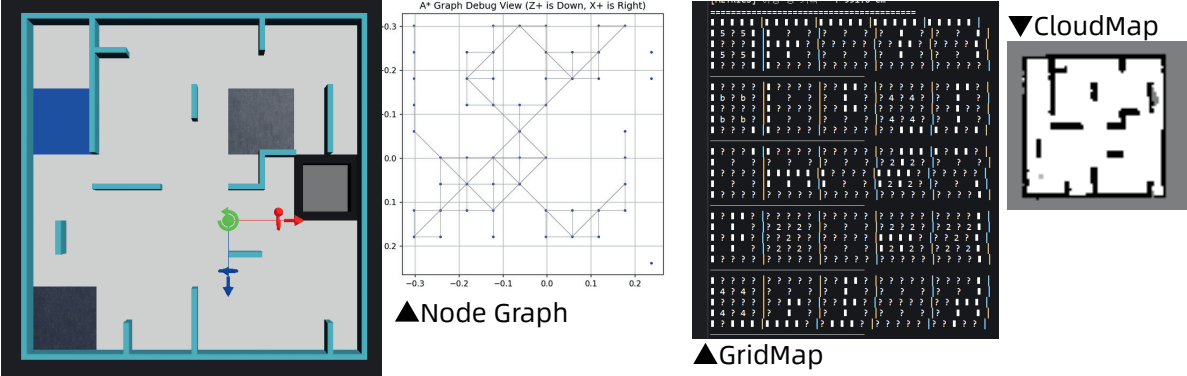
- Frontier Selection: Nodes adjacent to unvisited cells are designated as frontiers for exploration.
- Navigation: The A algorithm* is employed to calculate the optimal path to the target frontier.
- Cost-Aware Pathing: A 8x cost multiplier is applied to edges within swamp regions to discourage traversal and encourage path optimization around these areas.

Exception Handling & Safety

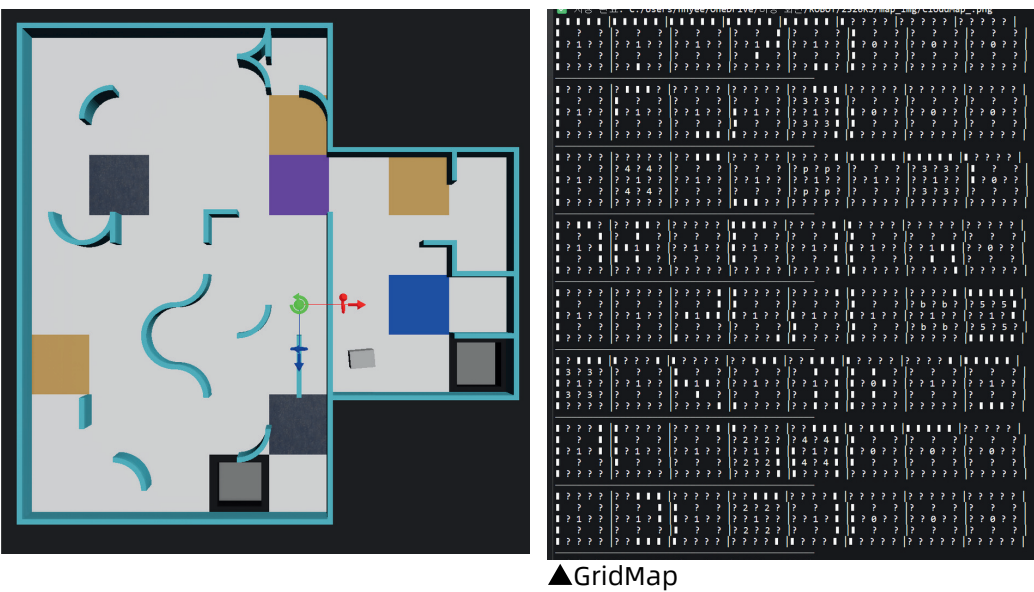
- Obstacle Avoidance: If an obstacle is detected at the same coordinate more than three times, the corresponding node is permanently pruned from the graph.
- Hazard Response: Upon detecting a hole, the robot triggers an immediate reverse maneuver, marks the area as impassable, and forbids further re-entry.

Termination Condition Exploration is concluded once all valid frontiers are exhausted. The robot then autonomously returns to the starting tile and transmits a completion signal.

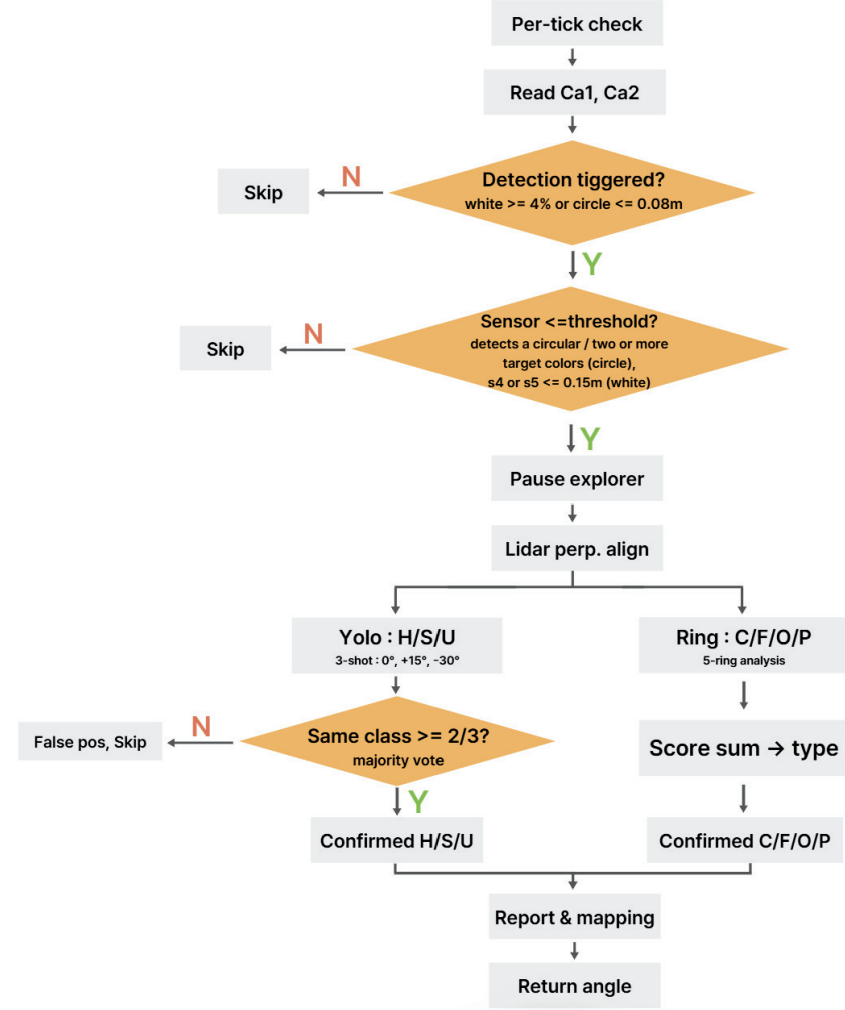
Self-made Half-Tile Wall Testing Map



Self-made Curved Wall Testing Map



WALL TOKEN DETECTION



Victim Detection Strategy

The robot detects two categories of victims using dual front-angled cameras (Ca1: left, Ca2: right).

Trigger Conditions

Detection is initiated when either camera satisfies one of the following:

- YOLO victims (H/S/U): White pixel ratio $\geq 4\%$ in camera image AND corresponding side distance sensor ≤ 0.15 m
- Circle victims (C/F/O/P): Concentric circle detected with estimated distance ≤ 0.08 m AND side distance sensor ≤ 0.08 m

Each tile is limited to a maximum of 2 detection attempts to prevent redundant processing.

YOLO-Based Detection (H / S / U)

Targets with white backgrounds (Harmed, Stable, Unresponsive) are classified via YOLOv5.

1. Advance toward victim wall (670 ms, center-maintaining PD control)
2. Align robot perpendicular to the wall using LiDAR linear regression on wall point cloud
3. Three-angle sampling: capture at 0° , $+15^\circ$, -30° relative to aligned angle
4. Run YOLOv5 inference (confidence threshold: 0.78, max detection distance: 0.25 m) at each angle
5. Majority vote: if the same class appears in ≥ 2 of 3 shots $\rightarrow$ confirmed victim; otherwise discarded as false positive

Concentric Circle Detection (C / F / O / P)

Victims marked with colored concentric rings are classified by ring color analysis.

1. Advance + LiDAR perpendicular alignment (same as above)
2. Fit ellipses to four color masks (Red, Green, Yellow, Blue) and determine outer radius
3. Divide the circle into 5 radial bands; sample 180 angles $\times$ 8 radii per band
4. Vote dominant color per ring; apply scoring: Red = -1, Yellow = 0, Green = +1, Blue = +2, Black = -2
5. Sum of 5 ring scores $\rightarrow$ classification: 0 = F, 1 = P, 2 = C, 3 = O

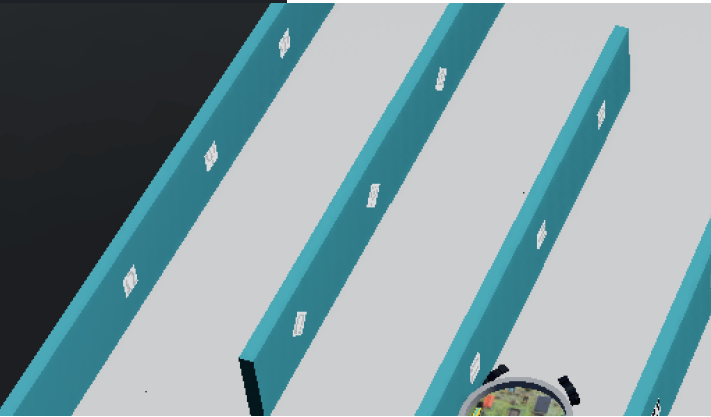
YOLOV5

```
300 epochs completed in 5.705 hours.
Optimizer stripped from /content/drive/MyDrive/2026 KOR/new6/finetune_v2/weights/last.pt, 14.5MB
Optimizer stripped from /content/drive/MyDrive/2026 KOR/new6/finetune_v2/weights/best.pt, 14.5MB

Validating /content/drive/MyDrive/2026 KOR/new6/finetune_v2/weights/best.pt
Fusing layers...
Model summary: 157 layers, 7023004 parameters, 0 gradients, 15.8 GFLOPs

Class Images Instances P R m
all 688 688 0.983 0.986 0
H 688 219 0.982 0.985 0
S 688 236 0.988 0.992 1
U 688 238 0.998 1 0

Results saved to /content/drive/MyDrive/2026 KOR/new6/finetune_v2
```



Victims are classified using a YOLOv5 object detection model trained on Webots screenshots and Greek letter datasets. Initial testing achieved 87.1% validation accuracy, but only 3 of 15 victims were detected in the simulation environment.

To improve performance, unlabeled images were removed, robot POV images were added, Adaptive Equalization replaced Histogram Equalization, and augmentation settings were optimized. After retraining, the model achieved 86.5% validation accuracy but successfully detected 15 of 15 victims in Webots.